Dr. Samantha Vance

Principal Software Architect and Senior ML Researcher

Education

Ph.D. in Computer Science, 2020 | B.Tech in Computer Science, 2015 — CGPA 9.8 / 10

Technical Skills

C++, Python, Java, SQL, Data Structures & Algorithms, Object-Oriented Design,
System Design, Operating Systems, Computer Networks, Machine Learning, scikit-learn,
Git, Docker, CI/CD, Linux, Redis, PostgreSQL, MySQL, REST APIs

Projects

• Scalable distributed pub-sub messaging broker with zero-copy network buffers in C++

• Recommendation engine using collaborative filtering and neural embeddings evaluated with precision/recall

• High-throughput database indexing engine supporting ACID transactions and query optimization

Experience

• Principal Architect at Tech Global — led distributed systems architecture migration to Kubernetes and microservices

• Research Scientist at AI Labs — published papers on logistic regression and gradient boosted decision trees

Hackathons / Internships

• Winner, Global AI Hackathon 2022

• Research Intern, Systems Lab — designed round-robin process scheduling simulators

Certifications

AWS Certified Solutions Architect

Achievements

Authored 5 IEEE papers on distributed database normalization and latency optimization

Preferred Roles

Principal Software Engineer, Chief Architect, Lead Data Scientist